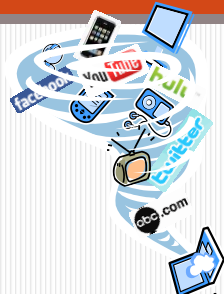


CM 7028 Digital Media Production



Chutisant Kerdvibulvech, Ph.D.
Associate Professor

รศ.ดร.ชุตินันต์ เกตวิบูลย์เวช

นศ. 7028 การผลิตสื่อดิจิทัล

L1

INSTRUCTOR

Assoc. Prof. Dr. Chutisant Kerdvibulvech

รศ.ดร.ชุตินันต์ เกตวิบูลย์เวช

- Email: chutisant.ker@nida.ac.th
- Website: www.gscm.nida.ac.th/~chutisant/
- Facebook: <https://www.facebook.com/chutisant.kerdvibulvech>
- Office: 8th Floor (GSCM)

(Please Email/Facebook me first to make sure that I will be there.)

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CM7028/CS-CM7028.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L2.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L3.pdf>
 - etc

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Evaluation

(Technique-based, Creativity-based, and Research-based)

- Project I / Assignments and Participation
30 %
- Project II / Assignments and Participation
30%
- Final Project / Assignments and Participation
30%
- Overall Class Participation and Attendance
10 %

Extra Points: 10% for Rinkou

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Textbooks

- Mixed Reality: Merging Real and Virtual Worlds, Yuichi Ohta, Hideyuki Tamura, 418 pages, Springer, ISBN-13: 978-3642875144 (April 15, 2014)
- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, 2nd Edition (2016)
- สื่อดิจิทัลใหม่...สื่อแห่งอนาคต, ชุดสันต์ เกิดวิบูลย์เวช คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ (นิด้า) 2560

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Related-Academic Papers

- C. Kerdvibulvech, "Vision and Virtual-based Human Computer Interaction Applications for a New Digital Media Visualization", 23rd International Conference on Computer Graphics, Visualization and Computer Vision 2015 (WSCG), Eurographics Association & ACM SIGGRAPH (Scopus, SCImago and Thomson Reuters ISI), Plzen, Czech Republic, June 8 - 12, 2015.
- C. Kerdvibulvech, "An Innovative Use of Multidisciplinary Applications Between Information Technology and Socially Digital Media for Connecting People", The Multidisciplinary International Social Networks Conference 2015 (Springer), in Communication of Computer and Information Science (CCIS), (Scopus and SCImago Quartile 4th), Ehime, Japan, September 1 - 3, 2015.

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Related-Academic Papers

- C. Kerdvibulvech, "A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models," EURASIP Journal on Embedded Systems (JES), 9 pages, No. 18, Springer Publisher (Elsevier's Scopus, EiCompendex, INSPEC, SciFinder, SCImago, ACM, DBLP, and DOAJ), 2014.
- C. Kerdvibulvech and K. Yamauchi, "Structural Human Shape Analysis for Modeling and Recognition," Joint IAPR International Workshops on Structural and Syntactic Pattern Recognition (S+SSPR'14), Lecture Notes in Computer Science (LNCS), vol. 8621, Joensuu, Finland, Springer-Verlag Berlin Heidelberg, pp. 282-290, August 20-22, 2014.
- เอกสารคำสอน นน 7028 การผลิตสื่อดิจิทัล คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ โดย ผศ.ดร.ชุดสันต์ เกิดวิบูลย์เวช, 2559.

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>
- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Overview

- Background
- Projects from related-topics of Digital Media

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Education



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of Information and Computer Science
Keio University, the oldest university in Japan

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Academic Work Experience

- Teaching and being as a research committee in following universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Mahidol University, Kasetsart University
 - NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Media Work Experience

Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



Assoc. Prof. Dr. Chutisant Kerdvibulvech

Digital Media Production

Projects

- Introduction (1 week)
- Project I (4 weeks)
 - Touch Communication (National University of Singapore)
- Project II (About 5 weeks)
 - Chatbot (Artificial Intelligence and Machine Learning)
 - Virtual Reality / Facebook 360 / Else
- Final Project (About 5 weeks)
 - Website and SEO (Search Engine Optimization)

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Digital Media

Digital media are any media that are encoded in a machine-readable format. Digital media can be created, viewed, distributed, modified and preserved on digital electronics devices.

["Digital Media"](#) (PDF). *Technology Brief*. University of Guelph. September 2006. Retrieved 28 March 2014.

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



aurora.dawn.com

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Future of Digital Media?



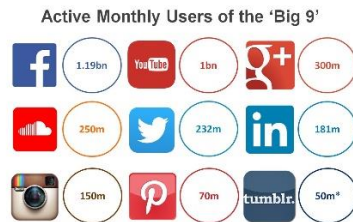
www.3dsg.com.au

- Big Data / SEO (Search Engine Optimization) / SEM (Search Engine Marketing)
- AI (Artificial Intelligence) / Deep Learning
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)
 - New Digital Media

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Digital Media Trends for 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
*Google+ data is estimated based on monthly user data and this figure is based on an estimate from
Facebook's own data. Facebook's data is based on an estimate from Facebook's own data.

kamber

::Source::
Kamber (Sydney, Australia)

kamber | 14 Social Media Trends for 2014
Assoc. Prof. Dr. Chutisant Kerdvibulvech

Kamber.com.au | 14

Digital Media Trends for 2016



Assoc. Prof. Dr. Chutisant Kerdvibulvech

<https://www.facebook.com/zuck>

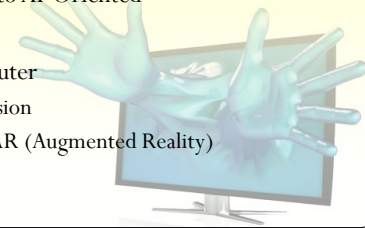
Digital Media Trends for 2017/2018



Assoc. Prof. Dr. Chutisant Kerdvibulvech

The Future of Digital Media

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- From Mobile-Oriented to AI-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



Assoc. Prof. Dr. Chutisant Kerdvibulvech

www.3dsg.com.au

Augmented Reality vs. Virtual Reality

Augmented Reality

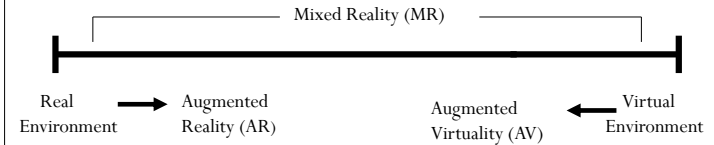
- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Miligram's Reality-Virtuality Continuum



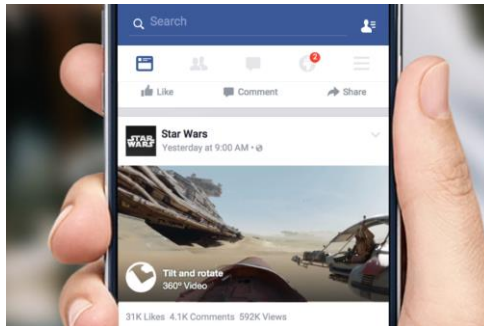
csis.pace.edu

Milgram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

Assoc. Prof. Dr. Chutisant Kerdvibulvech

From Virtual Reality to Facebook360



Assoc. Prof. Dr. Chutisant Kerdvibulvech

Virtual Reality



"The Oculus community is leading us into the future, and we can't wait for everyone to experience the magic of VR and the next platform."...**Mark Zuckerberg, CEO of Facebook**

Assoc. Prof. Dr. Chutisant Kerdvibulvech

24 September 2015

Chatbot

- A computer program which conducts a conversation via auditory or textual methods
- A chatter robot is a type of conversational agent, a computer program designed to simulate an intelligent conversation with one or more human users via auditory or textual methods
- AI (Artificial Intelligence) / Deep Learning

<https://en.wikipedia.org/wiki/Chatterbot>

Assoc. Prof. Dr. Chutisant Kerdvibulvech

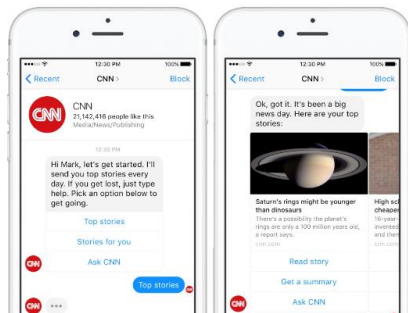
When Facebook understands our messages automatically..

เมื่อเฟซบุ๊กเข้าใจข้อความของเราได้เอง (1) (2) (3)..หนังสือพิมพ์เดลินิวส์

- <http://www.dailynews.co.th/it/502607>
- <http://repository.nida.ac.th/bitstream/handle/662723737/3148/59104.pdf?sequence=1&isAllowed=y>
- <http://repository.nida.ac.th/handle/662723737/3151>

Assoc. Prof. Dr. Chutisant Kerdvibulvech

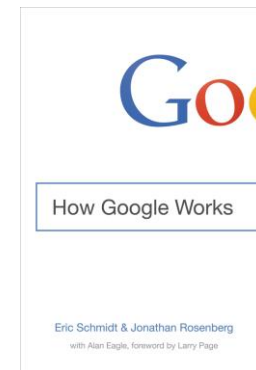
Chatbot by CNN



Assoc. Prof. Dr. Chutisant Kerdvibulvech

Rinkou: Extra Points 10%

- <https://www.amazon.com/How-Google-Works-Eric-Schmidt/dp/1455582344>



Assoc. Prof. Dr. Chutisant Kerdvibulvech

Thank you

Assoc. Prof. Dr. Chutisant Kerdvibulvech
Graduate School of Communication Arts and Management Innovation,
National Institute of Development Administration (NIDA)

chutisant.ker@nida.ac.th

<https://www.facebook.com/chutisant.kerdvibulvech>

Assoc. Prof. Dr. Chutisant Kerdvibulvech